

## Day 2 — The Carry Case

**Module:**  $\times 11$  Multiplication — Ekādhikena Pūrveṇa · **Band:** Senior · **Time:** 45 min

### Learning objective

By the end of this lesson, students will: compute  $87 \times 11$ ,  $65 \times 11$  correctly using the carry rule.

### Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

### Vocabulary introduced today

- *sūtra* (IAST: *sūtra*; lit. “aphorism / formula”)

### Lesson flow

#### Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

#### Core (20 min)

1. **Try  $87 \times 11$ .** Apply the rule:  $8 \_ (8+7) \_ 7 = 8 \_ 15 \_ 7$ . But 15 doesn’t fit in one slot. So *carry*: keep the 5 in the middle, add the 1 to the 8  $\rightarrow 9 \ 5 \ 7 = \mathbf{957}$ .
2. **General rule extension.** When the middle sum  $\geq 10$ :
  - Write the units of the sum in the middle.
  - Carry the tens digit to the left, adding it to the existing left digit.
3. **Practice progression.**
  - $65 \times 11 = 6 \_ (11) \_ 5 \rightarrow 715$
  - $49 \times 11 = 4 \_ (13) \_ 9 \rightarrow 539$
  - $99 \times 11 = 9 \_ (18) \_ 9 \rightarrow 1089$
  - $84 \times 11 = 9 \_ (12) \_ 4 \rightarrow 924$
4. **Cross-check with standard long multiplication.** Have students verify two of the answers using the traditional algorithm. “*Same answer? Good. The Vedic method is just a different procedure for the same arithmetic.*”
5. **Limit recognition.** “*When might this be slower than long multiplication?*” For numbers with three or more digits where carries cascade, the saving shrinks. Tomorrow we extend to 3-digit and meet the algebraic proof.

## Activity (15 min)

A scaled version of *Mental Math Sprint Cards* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces The Carry Case.

## Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

## Homework

(See [homework.md](#) for today’s task.)

## Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

## Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).